

Title: **How can *Agastya*'s programs be effectively implemented in municipal schools?**

Author: Nandini Talwar

Contact: +919619177974

Email: nandinikaurtalwar@gmail.com

“Economy is too serious a subject to be left to economists alone.”

-Jitendra Singh, Minister of State in Prime Minister Office

Abstract

In India, a large percentage of 15 million youth who enter the workforce annually lack employable skills, reducing overall productivity. Lack of interest and perceived value of education often cause low retention and dismal learning outcomes, which fuels the poverty cycle and contributes to the socio-economic divide.

The Agastya International Foundation operates a hands-on science education program, equipping students attending government schools in rural areas with tools that encourage leadership, confidence, creativity, and teamwork. Their model is embedded in the existing formal education system; it has based its products on the government syllabi to remain relevant and complement the school curricula. Agastya involves all critical stakeholders in education—children, teachers, parents, community, and government.

This research paper will address the question, “**How can *Agastya’s* programs be effectively implemented in municipal schools?**” after addressing the multiple hurdles in the funding of municipal schools in Mumbai that need to be overcome.

Agastya’s models are made to be scaled to different cities around India, and their credibility of their programs is being noted by certain state governments. Karnataka’s government, in partnership with Agastya, is replicating the Creativity Lab as part of its initiative to transform education. The governments of Bihar and Punjab have requested 100 mobile science labs and 22 mobile science labs and centers respectively. Given the current abysmal quality of education received in municipal schools run by the Brihanmumbai Municipal Corporation (BMC), there is a dire need for programs that are sustainable. Agastya’s models are created as such that there is no lack of motivation while delivering information in the classroom because they are hands-on and involve the students’ participation. Hence if Agastya’s models are implemented in BMC schools, the overall quality of education can increase, leading to a rise in the productivity of the future workforce.

Agastya's impact

In a report of Agastya published in 2015, it is estimated that more than 130 mobile labs spread across over a dozen states in India and 50 science centres spread across 18 states take Agastya's creative pedagogy to more than 1.5 million children and 200,000 teachers every year. Situated in Kuppam, Andhra Pradesh, Agastya's creativity lab is visited by more than 500 students per day, who learn using their hands-on teaching methods.

Agastya was founded in 1999 in the rural area of Kuppam, but in 2003 it worked on a scalable model to reach millions of rural children, teachers, and communities across India for widespread and sustainable impact. The following year, the State Government of Andhra Pradesh supported the expansion of Agastya's mobile labs into rural areas.

Initially, Agastya got teachers from villages around Kuppam because they understood the local customs, knew the potential areas of concern, and could teach in the local language. Now it holds teacher training programs across India, where teachers become part of Agastya's force, and they spread their impact by teaching using Agastya's science kits.

Some of these workshops are conducted on Agastya's Kuppam campus. Teachers from the states of Andhra Pradesh, Assam, Karnataka, Maharashtra, Tamil Nadu, and Telangana came to Agastya's Kuppam campus for a "Make Your Own Lab" (MYOL) workshop which was a residential workshop for duration of 2 to 4 days. Additionally, Agastya teachers also conduct training workshops in different cities around India. The teacher training MYOL program has oriented more than 350 teachers and cost incurred per Teacher varied from Rs. 12,000 to Rs. 16,000. (Over 5000 Government Teachers have been trained by Agastya in past three years by organizing two to four days of residential workshops)

A teacher training program conducted in August 2017 in Mumbai called "Make Your Own Lab" was a Maverick Teachers' Training Program budgeted at 6.54 lakhs. Teachers from BMC schools around Mumbai participated in the program, and a science kit was given to the teacher at the end of the program, which cost approximately 10,000 each. 20 - 23 teachers were present at the workshop which was operating from 8th - 11th August, 2017 from 9.30am to 5.30pm. Working in groups of 3-4 teachers, they were each given different science concepts and materials, and made to think of a creative way to explain that concept using those materials. 2-3 Agastya teachers were present at the workshop to moderate the proceedings and ensure that the concepts were explained in a correct manner.

Problems with the Indian schooling system

Over 95% of Indian children aged six to fourteen are enrolled in school, but even after attending five years of school, only 50% of all children in the fifth grade can read simple text suited to second grade children (Pratham Foundation, 2017). The Indian schooling system is structured by age and grade, and many students do not meet the academic standards set for them because they come from low-income Marathi speaking families that don't speak English. Weak foundational skills hinder the potential for progress in later years. This is demonstrated by a trend that shows flat learning trajectories based on empirical evidence that tracks children longitudinally.¹

Data indicates that enrolment rates for both boys and girls aged 6-14 was 96.9% in 2016 and for the age group 15-16 was 84.7% in 2016.²

Although enrolment has increased, the quality of the education received is decreasing. The Annual Status of Education Report (ASER) also notes that the proportion of children enrolled in Std II and Std IV but who are sitting with other grades has been increasing over time. In primary schools, in 2010, 55.2% of Std II classes sat with other grades compared to 63.7% in 2016. Similar trends are also visible for Std IV. The proportion of classes in which Std IV children are sitting with other grades increased from 49% in 2010 to 58% in 2016.³ When students don't sit in classes allotted to them, it is likely that they will not learn information suited to their knowledge base. When sitting in higher classes, students can get demotivated, and in lower classes, students can waste time re-learning the same concepts hence not developing new critical thinking skills. This data indicates that children can complete five years of their schooling without acquiring basic reading and math skills- the building blocks of primary education and the pathway to academic progress.

ASER data estimates that approximately 50 million children through Grades 3 to 5 need urgent help to complete elementary school effectively. Given the size of enrolment in Indian schools, the estimate annual education budget of the Brihanmumbai Municipal Corporation (BMC), and the size of the problem, solutions should be low cost, scalable to municipal schools in other cities, and rapidly functional.

The BMC's budget for 2015-16 was Rs 37,052 crores out of which Rs. 2,630 crores were allocated for 397,085 students. Per-child allocation for municipal school students has remained at Rs. 59,744 as per 2015-16 Budget estimates (excluding grants to private primary aided schools but including account revenue expenses and capital expenses/project works). After deduction of capital expenses from the budget, per child allocation amounts to Rs.50,534 in 2015-16. (Praja Foundation, 2016)

The outcome from education expenditure has had little to no change. The dropout rates have only increased, with 13 in every 100 students dropping out in 2014-15 in comparison to 4 in every 100 students in 2010-11. In English medium schools where enrolments have improved this year, there is a significant dropout of 5.3%. A time-series regression estimated that only 70.8% of students who enrolled in Class I in 2010-11 would make it to class 7 in 2016-17.

In the funds allocated for Education, capital expenditure has increased from Rs. 156.53 cr in 2014-15 to Rs. 357.14 cr in 2015-16, with no significant positive externality on the quality of education.

¹ Pritchett and Beatty (2012) The Negative Consequences of Overambitious Curricula in Developing countries. Working paper 293. Centre for Global Development. Washington DC.

² "The eleventh Annual Status of Education Report (ASER 2016)". *ASER*. 17 Jan. 2017. Web. 1 Nov. 2017. <http://img.asercentre.org/docs/Publications/ASER%20Reports/ASER%202016/aser2016_nationalpressrelease.pdf>

³ "The eleventh Annual Status of Education Report (ASER 2016)". *ASER*. 17 Jan. 2017. Web. 1 Nov. 2017. <http://img.asercentre.org/docs/Publications/ASER%20Reports/ASER%202016/aser2016_nationalpressrelease.pdf>

Importance of investment in Primary Education

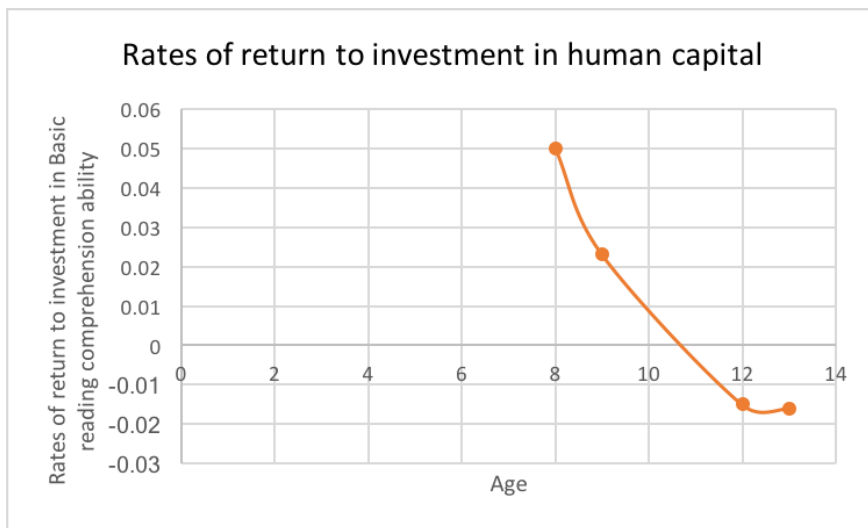
Blaug et al (1969), Tilak (1987), and Psacharopoulos (1993) show that investment in education yields a higher rate of return than investment in physical capital. Romer (1986) and Lucas (1988) have propounded the new growth theories in which sustained long-run growth of per capita income is explained by the likelihood of investment in human capital generating constant or increasing returns. (Ojha, Pradhan, 2006)

Hence for an effective solution in the long-run, the BMC's funds should be allocated towards increasing investment in Revenue expenditure, specifically to improve teacher training programs. In 2015-16, Rs. 17 crore was allocated for establishing 200 new computer labs in various municipal schools, which might have been a short-run solution, but in June 2015 an RTI query revealed that out of 1270 BMC-run schools in Mumbai, more than 75% did not have computer labs. This data indicates that the teachers are likely to not understand the workings of digital equipment and may not effectively deliver the information to their students. The policy implementation involved excessive bureaucracy and time lags, not to mention the time lag of 2-3 years involved in the transfer of knowledge to the teachers who will then transfer it to the students. The modern-day state of digital literacy still remains abysmal, proving that simply investing in computer labs will not make students digitally literate.

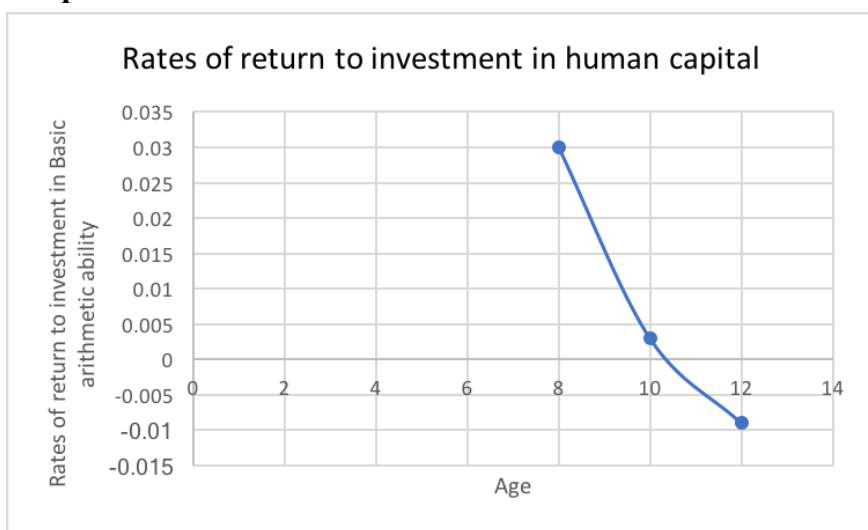
Hence there is a need for the programs implemented in BMC schools to place an emphasis on strengthening primary education through teacher training instead of simply investing in physical capital. Even while investing in human capital, the sector of primary education should be given most importance, as maximum results are yielded when investment in human capital is done during the early childhood stage, where brain growth is its fastest. Compared to later stages of childhood and education, the return from younger ages is the most (Heckman, 2007).

Hackman's hypothesis is demonstrated by the graphs below, where one of ASER's reports released in January 2017 had found that although reading ability had improved nationally, specifically in early grades in government schools, reading levels in Std VIII showed a slight decline since 2014. The same trend was seen for arithmetic, which showed improvement in government schools in primary grades, but among Std VIII students, it has continued to drop. This declining trend has been observed since 2010. The proportion of Std VIII students who could correctly do a 3-digit by 1-digit division problem was 68.4% in 2010. This number dropped to 44.2% in 2014, and has further declined to 43.3% in 2016.

Graph 1.0



Graph 1.1



Graphs 1.0 and 1.1 both demonstrate a trend that shows that as the age of the students increases, it becomes more and more difficult to train them to improve their basic reading and arithmetic skills. This is seen by the negatively sloped curve, which shows that the rate of return to investment in human capital decreases as age of the student increases.

Grade VII onwards the rate of return to investment becomes negative. In Graph 1.0, while moving from Grade VII to Grade VIII, there is a -0.1% rate of return, showing that the investment has not been effective enough and that the reading comprehension ability of Grade VIII students is of a lower standard in the year 2016 as compared to 2015. The same goes for basic arithmetic ability, in which Grade VII has experienced a -0.9% rate of return after investing in human capital like resources for teaching and teacher training to improve overall efficiency in imparting education and reducing wasted resources.

Hence to improve the quality of education, the BMC should know what to invest in to prevent allocative and productive inefficiency. Simply investing in human capital will not be effective- their needs to be investment in human capital more for primary education than secondary education owing to the extremely high opportunity cost faced if the primary knowledge base is not developed. Instead of spending more money on investing in education in higher grades, if the children have a considerably good base to begin with, the transition from grade to grade becomes much more feasible.

Agastya's programs

1. Mobile Science Labs

Agastya's outreach extends beyond the Kuppam campus to isolated rural villages where children find that travelling to the campus is too expensive and logistically impossible. Consequently, Mobile Labs were created as a cost-effective method to go into these villages and teach. Each mobile lab contains science models and low-cost experiments and is accompanied by two Agastya teachers.

Each MSL travels to doorstep of remote schools with 100+ hands-on science models covering a wide range of topics in Physics, Chemistry, Biology, and Math for classes 5 through 10. The teaching approach is consistent with the National Curriculum Framework and the topics overlap the NCERT syllabus. There are also experiments not directly correlated to the state syllabus but are intended to spark curiosity and encourage questioning among the children, as the demonstration sessions emphasize interaction and questioning. Children are able to relate better to concepts explained through interactive demonstrations rather than through rote learning, the method practiced in many municipal school. MSLs are highly effective in delivering large science fairs, and reaching community members through night community visits too. Night community visits allow the child beneficiary's parents, family and community members to participate in learning sessions about various beneficial topics.

The mobile science lab requires a fairly large vehicle like a van, 100 experiments and laptops. The capital expenditure for all the vans in 2016 was estimated to be approximately 18.17 lakhs, with one van costing approximately 2-3 lakhs. The operational expenditure was approximately 14.56 lakhs. A mobile lab can reach 9,000 to 10,000 children, for a total of more than 20,000 exposures, and 300 exposures for teachers. The total exposures for 2016 were 16,000. It expanded across the Taluk, district, and state levels. The number of exposures depends on the density of schools in the target catchment area.

The cost per exposure for one student per annum is estimated to be Rs. 204.5625. The cost may vary depending on the number of villages targeted, the productivity of teachers, and the literacy level of the villages which might reduce the time spent in each village and hence increase exposures.

2. iMobile Lab

Agastya's iMobile program focuses on teaching science concepts through science and computer labs. The program is intended for Class 7 students and targets 12-15 government schools a year. At the end of the academic year, literacy levels are expected to rise in science, language, and numeracy. Through its inclusive, participatory model, the program makes classroom-learning enjoyable.

The iMobile Lab is language neutral- it gives opportunities for students to improve both their vernacular and English skills. Since language is no barrier, the participation in the program has proven to be more active, as language lessons are not forced on the students.

3. Lab in a Box

Lab in a Box (LIB) is a unique concept designed by Agastya out of many years of experience to facilitate scale-up. Science models/experiments are organized concept-wise based on pre-designed learning modules, mapped to the school syllabus, and placed in boxes - typically 10 boxes covering majority of physics, chemistry, and biology concepts.

These boxes are then distributed to various schools across the target geographic area and a teacher from the government school is trained to use the models and experiments for hands - on teaching in the classroom. The boxes are left with the teacher for about a week and this facilitates hands - on learning in the school over a period beyond the visit of the Mobile Science Lab to that school. Agastya follows a rotation scheme of distribution involving 10 schools at a time, whereby the 10 boxes in one set of LIB are available to each of the ten schools for a week in a three - month period.

4. Lab on a Bike

To scale its reach and impact, Agastya had initiated the Lab in a Box (LIB) which has proved to be a great resource for government school teachers to execute hands - on science activities on a regular basis. The Lab on a Bike (LoB) program is a variation of the LIB program wherein an Agastya instructor uses the LIB to conduct hands - on science classes in the schools. The instructor travels to different schools each day along with the LIB box which fits onto his bike. On completion of all the pre - defined schools, he starts his next cycle with the second box and so on.

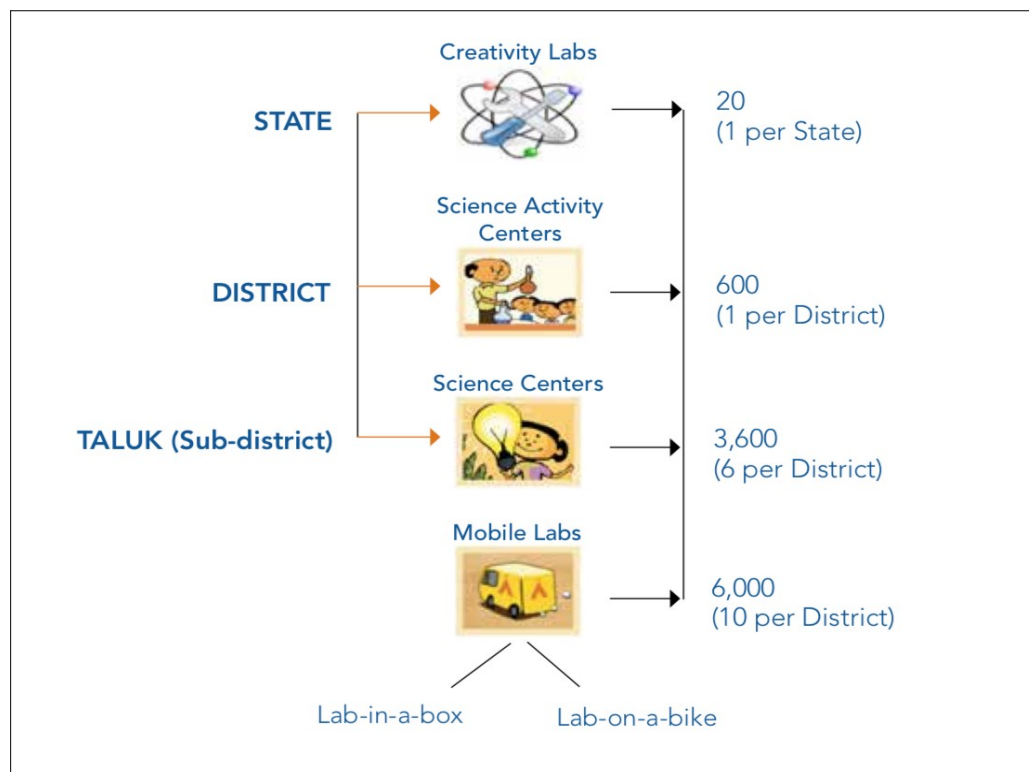
The LoB can cover about 10 schools per annum which will result in 10,000 + exposures with each child getting 10 exposures during the year.

5. Tech La Bike

TechLaBike is a revolutionary program that brings science education and digital literacy to the doorstep of a village school. An Agastya trained instructor reaches schools on a motorbike equipped with 'Lab in a Box' (LIB) and an Internet enabled laptop.

Laptops are used for simple mouse-based science games, quizzes, language neutral videos, and multimedia conferencing. The Agastya instructor travels to a set of 10 schools in one year, by visiting each school once to complete one cycle and then to reinforce the concepts that same year they travel to all the 10 schools again for another cycle.

The Tech La Bike is capable of 8000 exposures per year, and can reach a network of schools that are otherwise remote and inaccessible. Since equipment and digital resources are shared across multiple schools, the cost of the program is significantly reduced.



4

The following

table summarises Agastya's released data for 2016:

Program	Physical Requirements	Capex (Rs, lakhs)	Opex (Rs.Lakhs)	Exposures PA	Key Features	Benefits
Mobile Science Lab	Vehicle + 100 experiments	13.80	13.38	16,000	Mobile across Taluk, district and state.	High reach enables raising awareness of the importance of hands-on science on a mass level. Conducting programs such as Science Fairs and Community visits
iMobile Lab	Vehicle + 100 experiments + Laptops	18.17	14.56	12,000	Science + Technology for students and teachers across Taluk, district and state	High reach enables raising awareness of the importance of hands-on science on a mass level. Conducting programs such as Science Fairs and Community visits Digital literacy sessions
Science Fair	80+ Science Experiments	NA		500 - 8,000	Young Instructors demonstrate to their peers	Learning through participation, peer-to-peer teaching and hands-on activities
Lab on a Bike	Motorcycle + Science Equipment	2.25	4.34	10,000+	Lower cost mobile outreach program	Hands-on science activities on a regular basis in rural areas
TechLaBike	Motorcycle + Science Equipment + Laptop	3.17	4.34	8,000	Science + Technology for students and teachers	Low cost model providing multimedia and hands-on science sessions

⁴ World Bank. "Agastya: Improving Critical Thinking and Leadership Skills". Innovationpolicyplatform.org. n.d. Web. 1 Nov. 2017.

<https://www.innovationpolicyplatform.org/system/files/Agastya%20case%20study_April20%202017.pdf>

From the above data, it is evident that the TechLa Bike (compared to the Lab on a Bike) is the most effective program implemented to reach out to rural villages. Although the exposures for the Lab on a Bike exceed 10,000 and the TechLa Bike stands at 8000, the positive externality of digital literacy achieved from the TechLa Bike is increasingly valuable and is required in municipal schools.

The cost per exposure for the TechLa Bike is Rs. 93.875 per person and the cost per exposure for the Lab on a Bike is Rs. 65.9 per person. The Lab on a Bike is more effective for students in rural areas with no access to computers, but for municipal schools, the TechLa Bike program is more effective as it provides technology training to students and teachers at a low cost, and the teachers in these municipal schools can continue reinforcing the learned concepts to increase the effectiveness of primary training. This has made it more feasible for students in rural areas impacted by Agastya to pursue employment related to the use of computers.

The iMobile Lab costing 32.73 lakhs, although more costly than the Mobile Science Lab costing Rs. 27.18 lakhs, includes the use of laptops required for digital literacy. The cost per exposure for the iMobile Lab is Rs. 272.75 per person whereas for the Mobile Science Lab stands at Rs. 169.875 per person.

There is a difference in cost per exposure of Rs. 102.75 per person, hence the Mobile Science Lab may be the more feasible program to implement in municipal schools that are experiencing a large dropout rate. The opportunity cost of the lost exposures is more than the opportunity cost of lost opportunities for increasing digital literacy. The laptops also require increased investment, and setting up computers for continuous reinforcement of learned concepts in municipal schools would include a time lag because of excessive bureaucracy.

Conclusion

Agastya's programs have proven to be effective in the rural areas of Kuppam, as the "Young Instructor Leaders" (YIL) which includes students who show great academic and leadership potential, have been successful in pursuing their path to higher education. Agastya reports that 6,000 graduates from the YIL program are now active agents of social change, 26 YILs have reached the finals of the national-level science competition; 11 have won awards for five years successively. 30 YILs have received scholarships to colleges.⁵

Yearly series of evaluation studies have helped track Agastya's progress, and shown the effectiveness of their policies. Agastya's hands-on learning kits show that just distributing teaching-learning materials does not have any effect on the quality of education received, but having the students actively involved in this by training teachers to do so does.

The "MYOL" program implemented in Mumbai, based on all the participants' feedback, case stories, and informal interviews substantiated that the program was very effective (qualitatively). It was successful in motivating the teachers that teaching-aids can be prepared using low cost materials. Short periods of teacher training programs (that last 2-3 days) are effective in reinforcing concepts in teachers, in addition to sitting in small groups, which can lead to better teaching methods. For teacher training, the MYOL workshops should be implemented, which will improve the quality of education received by students in municipal schools.

However, implementing only teacher training programs is a short-term solution. In the long run, for the quality of education to improve, 2-3 teachers from every municipal school in India should be trained by Agastya teachers and become part of the Agastya teaching force so that they can refresh concepts learned in workshops on a monthly basis. If we train more and more teachers in municipal schools to train other teachers in the same vicinity, the program can have a ripple effect. This can be the fastest method for increasing the standard of education in municipal schools without compromising quality.

⁵ World Bank. "Agastya: Improving Critical Thinking and Leadership Skills". Innovationpolicyplatform.org. n.d. Web. 1 Nov. 2017.
<https://www.innovationpolicyplatform.org/system/files/Agastya%20case%20study_April20%202017.pdf>